#ManyDaughters Project: Pre-Analysis Plan (PAP)

Authors: (11 Jul 2025)

A) Common Design Choices (Apply to all four hypotheses)

Item	Decision & rationale
Data source & waves	SOEP-Core release v37; individual questionnaire waves 2018 & 2019 (latest two pre-COVID years with harmonised wording).
Unit of observation	One row per *respondent – year* (pid x syear)
Sample filters	• Private households only (pop ∈ {1,2}) • Respondents aged **≥ 40** in the survey year (≈ completed fertility) • Observations retained only if the outcome and treatment are non-missing after recoding.
Treatment variable	daughters = count of daughters across kidsex01–kidsex19. (For interpretation we also speak of "having daughters".)
Control variables	Age & square, total number of children (i.kids dummies), log household net income (lnhhinc from hghinc), years of schooling (pgbilzeit), employment status dummies (pgemplst), migration background (migrant), East Germany dummy (east), survey-year FE (i.syear).
Estimation	OLS with survey weights [pw = phrf]; standard errors clustered at respondent level (cluster(pid)).
Statistics reported	Coefficient, SE, t-statistic, p-value, DF, N, # participants, and a textual directionality flag, exported to <hyp>_results.csv as specified in the task book.

B) H1: Daughters -> Support for gender equality (attitudes)

Section	Plan
Outcome (X)	geq_index: mean of five 7-point items (plj0621, plj0622, plj0623, plj0624, plj0625) re-scaled so higher = more egalitarian, then z-standardised.
Key regression	reg geq_index daughters [covariates], cluster(pid) pw(phrf)
Hypothesis test	Null H0: β_daughters = 0; alternative H1: β_daughters > 0. Significance judged by two-sided t-test (α = 0.05). A positive and significant β_daughters ⇒ hypothesis supported.
Interpreation rule	Directionality flag = "in line with hypothesis" if $\hat{\beta}$_daughters > 0.

C) H2: Daughters -> Left-leaning political preferences

Section	Plan
Outcome (Y)	z_left: respondent's 0-10 left-right self-placement (plh0004), reversed (10 - = most left) and z-standardised.
Key regression	reg z_left daughters [covariates], cluster(pid) pw(phrf)
Hypothesis test	Null H0: β_daughters = 0; alternative H1: β_daughters > 0. Significance judged by two-sided t-test (α = 0.05). A positive and significant β_daughters $\Rightarrow$ hypothesis supported.
Interpreation rule	Directionality flag = "in line with hypothesis" if $\hat{\beta}$_daughters > 0.

D) H3: Daughters -> Household gender equality

Section	Plan
Outcome (Z)	z_eq: 7-point agreement with "Best if man and woman work the same amount so they can share responsibility" (plh0308), z-standardised (higher = more equality oriented).
Key regression	reg z_eq daughters [covariates], cluster(pid) pw(phrf)
Hypothesis test	Null H0: β_daughters = 0; alternative H1: β_daughters > 0. Significance judged by two-sided t-test (α = 0.05). A positive and significant β_daughters ⇒ hypothesis supported.
Interpreation rule	Directionality flag = "in line with hypothesis" if $\hat{\beta}$_daughters > 0.

E) H4: Daughter effect stronger for men

Section	Plan
Outcome (X)	Same geq_index as H1.
Additional variables	male = 1 if respondent's sex == male; interaction D_x_M = daugthers x male.
Key regression	reg geq_index daughters male D_x_M [covariates], cluster(pid) pw(phrf)
Hypothesis test	Null H0: β_D_x_M = 0; alternative H1: β_D_x_M > 0. Significance judged by two-sided t-test (α = 0.05). A positive and significant β_D_x_M ⇒ hypothesis supported.
Interpreation rule	Directionality flag = "in line with hypothesis" if $\hat{\beta}$_D_x_M > 0.

F) Notes on coding & exclusions (all hypotheses)

- **Recoding scheme for value labels**
  - SOEP non-substantive codes -1, -2, -3, -5, -8 are set to missing (.).
  - Reverse-coding: rev = max – raw.
  - Standardisation: egen z_var = std (var).

- **Weighting & clustering** follow SOEP best practice for cross-sectional analyses.